Fall 2001

Problem 1 (Fa01). Let A be an $n \times n$ matrix with real entries, let $\chi(t)$ denote its characteristic polynomial, and let $g(t) \in \mathbb{R}[t]$ be a polynomial of degree $n-1$ dividing $\chi(t)$. What are the possibilities for the rank of $g(A)$?

Problem 2 (Fa01). Let the function $f : \mathbb{R} \to \mathbb{R}$ be bounded on bounded sets and have the property that $f^{-1}(K)$ is closed whenever K is compact. Prove f is continuous.

Problem 3 (Fa01). Find all finite abelian groups G, up to isomorphism, such that the group of automorphisms of G has odd order.

Problem 4 (Fa01). Let $a = \dfrac{1+i}{2}$, $b = \dfrac{-1+i}{2}$. Let Γ be the polygonal path in the complex plane with successive vertices $-1, -1+i, 1, 1+i, -1$. Evaluate the integral

$$I = \int_\Gamma \frac{1}{(z-a)^2(z-b)^3}\,\mathrm{d}z$$

Problem 5 (Fa01). Let ε be a positive number. Prove that the polynomial $p(z) = \varepsilon z^3 - z^2 - 1$ has exactly two zeros in the half-plane $\Re z < 0$.

Problem 6 (Fa01). Let A be a symmetric $n \times n$ matrix over $\mathbb{R}$ of rank $n-1$. Prove there is a k in $\{1, 2, \ldots, n\}$ such that the matrix resulting from deletion of the k-th row and k-th column from A has rank $n-1$.

Problem 7 (Fa01). Let the function u on $\mathbb{R}^2$ be harmonic, not identically 0, and homogeneous of degree d, where $d > 0$. The homogeneity condition means that $u(tx, ty) = t^d u(x, y)$ for $t > 0$. Prove that d is an integer.

Problem 8 (Fa01). Which of the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 occur as the last digit of n^n for infinitely many positive integers n?

Problem 9 (Fa01). Let the nonconstant entire function f satisfy the conditions

- $f(0) = 0$;
- for each $M > 0$, the set $\{z \mid |f(z)| < M\}$ is connected.

Prove that $f(z) = cz^n$ for some constant c and positive integer n.

Problem 10 (Fa01). Prove that a commutative $\mathbb{C}$–algebra of 2×2 complex matrices has dimension at most 2 over $\mathbb{C}$.

Problem 11 (Fa01). Let S be the set of continuous real valued functions on $[0, 1]$ such that $f(x)$ is rational whenever x is rational. Prove that S is uncountable.

Problem 12 (Fa01). Let $\mathbb{F}$ be a field, and let G be the group of 2×2 upper triangular matrices over $\mathbb{F}$ of determinant 1.

1. Determine the commutator subgroup of G.
2. Suppose $\mathbb{F}$ is the finite field of order p^k. Determine for this case the minimum number of generators for the subgroup found in Part 1.

Problem 13 (Fa01). Let the entire function f be real valued on the lines $\Im z = 0$ and $\Im z = \pi$. Prove that f has $2\pi i$ as a period: $f(z + 2\pi i) = f(z)$ for all z.

Problem 14 (Fa01). Let F be a polynomial over $\mathbb{C}$ of positive degree d, and let S be its zero set. Prove that every rational function R whose finite poles lie in S can be written uniquely as

$$R = \sum_{k=m}^{n} a_k \, F^k$$

where m and n are integers, $m \leqslant n$, and the a_k are polynomials whose degrees are less than d.

Problem 15 (Fa01). Let A be an $n \times n$ complex matrix such that the three matrices $A + I$, $A^2 + I$, $A^3 + I$ are all unitary. Prove that A is the zero matrix.

Problem 16 (Fa01). Consider the second order linear differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + p(x)\frac{\mathrm{d}x}{\mathrm{d}t} + q(x)x = 0$$

Here $p(x)$ and $q(x)$ are continuous functions on $\mathbb{R}$. Assume that the solutions, defined for all t, are *translation invariant*, that is, if f is a solution and s is a real number, then the function $g(t) = f(t + s)$ is also a solution. Prove that p and q are constant.

Problem 17 (Fa01). Let $\mathbb{K}$ be a field. For what pairs of positive integers (a, b) is the subring $\mathbb{K}[t^a, t^b]$ of $\mathbb{K}[t]$ a unique factorization domain?

Problem 18 (Fa01). Evaluate the integrals $F(t) = \displaystyle\int_{-\infty}^{\infty} \frac{e^{-itx}}{(x+i)^3}\, \mathrm{d}x$ for values $-\infty < t < \infty$.